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&

Aptitude Test (Mathematics and Soft Skill)

Prepared by:

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Subject wise analysis

Subject	Number of questions
Basics of Science and Engineering (DDCET)	50
Aptitude Test (Mathematics and Soft Skill) DDCET	50
Total	100

Q1. Burning of coal is a which type of process ?

- A. Double displacement
- B. Decomposition
- C. Combination
- D. Displacement

Correct Ans: C

Solution :

⇒ Burning of coal involves the main component, carbon, reacting with oxygen (O_2) in the form of carbon dioxide (CO_2).

⇒ This is a combination reaction where two elements or compounds combine to form a single substance.

Q2. $Na_2SO_{4(aq)} + BaCl_{2(aq)} \rightarrow BaSO_{4(aq)} + NaCl_{(aq)}$. Which type of reaction is this?

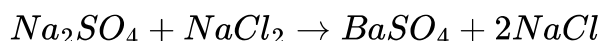
- A. Combination
- B. Double displacement
- C. Decomposition
- D. None of above

Correct Ans: B

Solution :

⇒ Double displacement reactions are reactions in which the reactants undergo an exchange of ions.

⇒ The example of a double displacement reaction is given below:



⇒ Here Cl_2 and SO_4 undergo exchange.

⇒ This equation can be expressed in a generalized form as follows: $AC + BD \rightarrow AD +$

Q3. Reaction of Zinc with Sulphuric acid gives Zinc Sulphate and Hydrogen. Which is the balanced equation for this chemical change?

- A. $Zn + 2H_2SO_4 \rightarrow 2ZnSO_4 + H_2$
B. $2Zn + H_2SO_4 \rightarrow ZnSO_4 + 2H_2$
C. $Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$
D. None of above

Correct Ans: C

Solution :

⇒ In the balanced equation, the quantity of atoms for each element stays constant, both and after a chemical reaction.

⇒ Here in option C, zinc (Zn) has 1 atom, hydrogen H has 2 atoms, sulphur S has 1 at oxygen O has 4 atoms on both the side. Hence that equation is balanced.

Q4. The outer layer of the teeth is made-up of

- A. Calcium nitrate
B. Potassium phosphate
C. Calcium chloride
D.) Calcium phosphate

Correct Ans: D

Solution :

⇒ Tooth enamel acts as a highly durable shield for your teeth, composed of a tough sub called calcium phosphate that is even stronger than bone.

⇒ This protective layer covers the visible part of your teeth

Q5. The pH of aqueous solution of which salt is 7.0 ?

- A. KNO_3
B. NH_4Cl
C. CH_3COONa
D. Na_2CO_3

Correct Ans: A

Solution :

⇒ KNO_3 is a salt formed by a strong acid (HNO_3) and a strong base (KOH).

⇒ When strong acids and strong bases react, they tend to neutralize each other, resulting solution with a pH close to 7.

Q6. Metal oxide + Acid → _____

- A. Base + Water
B. Salt+ Water
C. Acid + Salt
D. None of these

Correct Ans: B

Solution :

⇒ Metal oxides have a basic nature, so when an acid interacts with a metal oxide, they react with each other out.
⇒ This results in the formation of salt and water during the reaction.

Q7. State the molecular formula of Sodium Zincate.

- A. Na_2ZnO_3
- B. Na_2ZnO_2
- C. $NaZnO_2$
- D. $Na_2Zn_2O_2$

Correct Ans: B

Solution :

⇒ Sodium zincate is a chemical compound with the formula Na_2ZnO_2 .
⇒ It is formed when zinc reacts with sodium hydroxide ($NaOH$) in an aqueous solution.
⇒ It is often used in galvanizing and electroplating processes and can also be found in some types of batteries.

Q8. How many electrons are present in the outermost shell of fluorine?

- A. 9
- B. 7
- C. 2
- D. 5

Correct Ans: B

Solution :

⇒ The atomic number is 9. So it has 2 electrons in the innermost shell and 7 electrons in its outer shell.

Q9. Which of the following methods is suitable for preventing an iron frying pan from rusting?

- A. Applying grease
- B. Applying paint
- C. Applying a coating of zinc
- D. All of the above.

Correct Ans: C

Solution :

⇒ Applying a layer of zinc on an iron frying pan can prevent it from rusting.
⇒ This technique is called Electroplating.
⇒ Since zinc is more reactive than iron, it reacts with oxygen in the air first, keeping the iron intact.

Q10. Which of following metals is highly malleable?

- A. Gold
- B. Lead
- C. Both (A) and (B)
- D. None of these

Correct Ans: A

Solution :

⇒ Malleability defines material's capacity to undergo permanent deformation under compression without breaking.
⇒ Gold has an atomic structure that enables it to be hammered into extremely thin sheets which are so delicate that they appear almost transparent.
⇒ This is commonly referred to as gold leaf.

Q11. _____ is not valid data type in MS-Excel.

- A. Number
- B. Character
- C. Label
- D. Date/time

Correct Ans: C

Solution :

Note - This question is canceled by the authority.
⇒ In Excel, "label" is a word that's used for text entries that usually act as headers or identifiers in a spreadsheet, but it's not seen as a separate data type.
⇒ "Character" is also not visible directly in the MS Excel data type. It is included in "text" data type.

Q12. What is the max zoom percentage in MS-PowerPoint?

- A. 400
- B. 300
- C. 200
- D. 100

Correct Ans: A

Solution :

⇒ Powerpoint can zoom in up to 400% and zoom out to a minimum of 10%.

Q13. Which is not in MS-Word ?

- A. Italic
- B. Magic tool
- C. Font
- D. Bold

Correct Ans: B

Solution :

- ⇒ There is no such tool named Magic tool in MS-Word.
 - ⇒ While, font, bold, italic are the tools available to change the font styles.
-

Q14. CPU consist of

- A. ALU + CU
- B. ROM + ALU
- C. RAM + ROM
- D. None of the above

Correct Ans: A

Solution :

- The CPU (Central Processing Unit) typically consists of two main components:
- ⇒ ALU (Arithmetic Logic Unit): Responsible for performing arithmetic and logical operations on data.
 - ⇒ CU (Control Unit): Manages the execution of instructions and coordinates the activities of the other hardware components.
-

Q15. Which of the following is not OS (Operating System)

- A. UNIX
- B. MS-DOS
- C. LINUX
- D. Microsoft Office

Correct Ans: D

Solution :

- ⇒ Microsoft Office is not an operating system (OS).
 - ⇒ It is a suite of productivity software applications developed by Microsoft for use in offices and businesses.
-

Q16. Which pollutant is commonly associated with "blue baby Syndrome" in infants?

- A. Lead
- B. Arsenic
- C. Mercury
- D. Nitrate

Correct Ans: D

Solution :

- ⇒ Blue baby syndrome is often linked to nitrate pollution.
⇒ Nitrate can pollute sources of drinking water, and when babies drink water that has high levels of nitrate, it can cause a condition where the blood's ability to carry oxygen is decreased.
⇒ This can result in the skin turning a bluish color.
-

Q17. Clouds are present in _____ layers of atmosphere?

- A. Stratosphere
- B. Troposphere
- C. Mesosphere
- D. Thermosphere

Correct Ans: B

Solution :

⇒ Clouds are primarily present in the troposphere, which is the lowest layer of the Earth's atmosphere.

Q18. CFCs are not recommended to be used in refrigerators because they _____

- A. Increase temperature
- B. Affect environment
- C. Deplete ozone
- D. Affect human body

Correct Ans: C

Solution :

- ⇒ CFCs release chlorine atoms when they are broken down by ultraviolet (UV) radiation in the stratosphere.
⇒ These chlorine atoms then react with ozone (O_3) molecules, causing the depletion of the ozone layer.
⇒ The ozone layer is crucial for protecting life on Earth from harmful UV radiation.
-

Q19. What is the approximate efficiency range of modern solar photovoltaic (PV) panels in converting sunlight into electricity?

- A. 5- 10%
- B. 15 -20%
- C. 20 -30%
- D. 40 - 50%

Correct Ans: B

Solution :

⇒ The approximate efficiency range of modern solar photovoltaic (PV) panels in converting sunlight into electricity varies, but it typically falls between 15% to 20%.

⇒ However, some high-efficiency panels can achieve efficiencies upwards of 20% or even higher.

Q20. Which pollutant is primarily responsible for causing respiratory problems and cardiovascular diseases in humans?

- A. Lead
- B. Benzene
- C. Particulate matter (PM₁₀)
- D. Carbon Monoxide (CO)

Correct Ans: C

Solution :

⇒ Particulate matter (PM₁₀) is primarily responsible for causing respiratory problems and cardiovascular diseases in humans.

⇒ PM₁₀ refers to inhalable particles with diameters that are generally 10 micrometers or smaller.

Q21. What is the S.I unit of power?

- A. Joule
- B. Volt
- C. Watt
- D. Ampere

Correct Ans: C

Solution :

⇒ The SI unit of power is the watt (W) which is equal to one joule per second.

Q22. The pitch of micrometer screw is 0.5 mm. There are 50 divisions on its circular scale. What is the least count of the screw?

- A. 0.1 mm
- B. 0.001 mm
- C. 1 mm
- D. 0.01 mm

Correct Ans: D

Solution :

Given data:

⇒ Pitch distance of screw = 0.5 mm

⇒ Number of total divisions on circular scale = 50

$$\Rightarrow LC = \frac{\text{pitch distance of screw}}{\text{number of total division on circular scale}}$$

$$\Rightarrow LC = \frac{0.5}{50} = 0.01 \text{ mm}$$

Q23. 10 dyne= _____ N?

- A. 10^{-5}
- B. 10^{-4}
- C. 10^5
- D. 10^7

Correct Ans: B

Solution :

Relation between newton (N) and dyne:

$$1 \text{ N} = 10^5 \text{ dyne}$$

$$1 \text{ dyne} = 10^{-5} \text{ N}$$

$$10 \text{ dyne} = 10 \times 10^{-5} \text{ N} = 10^{-4} \text{ N}$$

Q24. What is the unit of Energy in CGS.

- A. Newton
- B. Watt
- C. Joule
- D. Arg

Correct Ans: The correct answer does not align perfectly with the provided option

Solution :

$\Rightarrow$ In the CGS (Centimeter-Gram-Second) system, the unit of energy is the erg. The erg is a small unit of energy.

$\Rightarrow$ In the MKS (Meter-Kilogram-Second) system and SI (International System of Units), the unit of energy is the joule.

$\Rightarrow$ The relationship between the erg and the joule:

$$1 \text{ erg} = 10^{-7} \text{ joules} \implies 1 \text{ joule} = 10^7 \text{ erg}$$

Q25. The refractive index of water is found to have the values 1.30, 1.32, 1.34, 1.35, 1.31. Calculate the mean absolute error.

- A. 0.01254
- B. 0.1254
- C. 1.254
- D. 0.001254

Correct Ans: The correct answer does not align perfectly with the provided option
Solution :

⇒ Standard value of refractive index of water = 1.333

Absolute differences:

$$|1.30 - 1.333| = 0.033$$

$$|1.32 - 1.333| = 0.013$$

$$|1.34 - 1.333| = 0.007$$

$$|1.35 - 1.333| = 0.017$$

$$|1.31 - 1.333| = 0.023$$

$$\text{Mean Absolute Error} = (0.033 + 0.013 + 0.007 + 0.017 + 0.023) / 5 = 0.093 / 5 = 0.0186$$

Q26. Which instrument is used to measure "Inner diameter of the ring"?

- A. Vernier caliper
- B. Micrometer screw
- C. Spherometer
- D. Physical balance

Correct Ans: A

Solution :

⇒ Vernier caliper is used to measure "Inner diameter of the ring."

Q27. The mathematical expression for Newton's second law is _____

- A. $F = mv$
- B. $P = mv$
- C. $F = ma$
- D. $P = ma$

Correct Ans: C

Solution :

Newton's second law of motion:

⇒ Newton's second law of motion states that the acceleration of an object is directly proportional to the net force acting on it and inversely proportional to its mass.

⇒ Formula: $F = m \times a$

Q28. Find the acceleration produced by applying a force of 30 N on an object of mass of 5 kg .

- A. 0.6 m/s^2
- B. 6 m/s^2
- C. 6 m/s
- D. 0.6 m/s

Correct Ans: B

Solution :

Given data:

$$\Rightarrow F = 30\text{ N}$$

$$\Rightarrow m = 5\text{ kg}$$

Force acting on the object $F = m \times a$

$$a = F/m$$

$$a = 30/5 = 6\text{ m/s}^2$$

Q29. _____ are the examples of impulse of force.

- A. China wares are wrapped in paper while packing.
- B. Bogies of a train are provided with the buffers.
- C. Both (A) and (B)
- D. None of above

Correct Ans: C

Solution :

Impulse of Force:

$\Rightarrow$ Impulse of force is the change in momentum of an object and it depends on the force and the time duration of the force.

$\Rightarrow$ Formula: Impulse of Force = force $\times$ time

$\Rightarrow$ SI unit: newton-second (N s)

$\Rightarrow$ CGS unit: dyne-second.

$\Rightarrow$ It is a vector quantity.

Examples:

1. China wares wrapped in paper while packing.
2. Bogies of a train are provided with buffers.

Therefore, from the above explanation the correct option is (C).

Q30. A body is moving in a vertical circular motion, which one of the following forces does not experience?

- A. Force of gravity
- B. Centripetal force
- C. Friction force
- D. Centrifugal force

Correct Ans: C

Solution :

⇒ A body moving in vertical circular motion experiences these three forces like Force of gravity, Centripetal force and Centrifugal force.

⇒ A body moving in vertical circular motion does not experience Friction force.

Q31. _____ is the field force.

- A. Gravitational force
- B. Magnetic force
- C. Electrical force
- D. All of above

Correct Ans: D

Solution :

⇒ Field forces are any forces that act on an object without direct physical contact.

⇒ There are three main types: 1. Gravitational force, 2. Magnetic force, 3. Electrical force.

Therefore, from the above explanation the correct option is (D) All of above.

Q32. The work done by the external forces on a system equals the changes in _____

- A. Total energy
- B. Kinetic energy
- C. Potential energy
- D. None of these

Correct Ans: A

Solution :

⇒ The work done by the external force on a system equals the change in total energy.

Q33. 6 V battery is connected across a resistance R . The current through R is 0.4 mA. What is the value of R ?

- A. 10 k Ω
- B. 15 k Ω
- C. 20 k Ω
- D. 24 k Ω

Correct Ans: B

Solution :

Given data:

$$\Rightarrow V = 6 \text{ V}$$

$$\Rightarrow I = 0.4 \text{ mA} = 0.4 \times 10^{-3} \text{ A}$$

$$V = I \times R$$

$$6 = 0.4 \times 10^{-3} \times R \Rightarrow R = 15000 \Omega = 15 \text{ k}\Omega$$

Q34. _____ is the Unit of permittivity.

- A. Nm²/C²
- B. A/m
- C. N/m²C
- D. C²/Nm²

Correct Ans: D

Solution :

According to Coulomb's law

The magnitude of the electrostatic force (F) between two point charges

$$F = k \frac{Q_1 Q_2}{d^2}$$

$$F = \frac{1}{4\pi\epsilon_0} \frac{Q_1 Q_2}{d^2}$$

$$\epsilon_0 = \frac{Q_1 Q_2}{4\pi \times F \times d^2} = \frac{C^2}{Nm^2}$$

Hence, option (D) is correct.

Q35. The Electric flux depends on the _____

- A. Electric field intensity
- B. Area
- C. Angle between the surface and the electric field.

D. All of these

Correct Ans: D

Solution :

⇒ The Electric flux depends on the Electric field intensity, Area and Angle between the and the electric field.

Electric flux (ϕ) is a measure of the electric field passing through an area.

Calculation: $\phi = E \times A \cos \theta$

Where,

E - Magnitude of the electric field intensity

A - Area of the surface

θ - Angle between the surface and the electric field.

Therefore, from the above explanation the correct option is (D) All of these.

Q36. Inverse of Resistance is known as _____

- A. current
- B. specific resistance
- C. conductance
- D. voltage

Correct Ans: C

Solution :

Resistance :

⇒ Resistance of the conductor is the opposition to the flow of current. It is denoted by R

⇒ Unit : Ω

Conductance :

⇒ Conductance is the offered by the conductor to the flow of current. It is denoted by G

⇒ Unit : $\mathcal{U}$ or S (siemens)

Hence, Inverse of Resistance is known as conductance means Conductance is the reciprocal of resistance ($G = \frac{1}{R}$)

Q37. What is the effective value of resistance when two resistors $10\ \Omega$ and $20\ \Omega$ are connected in parallel combination?

- A. $10\ \Omega$
- B. $20\ \Omega$
- C. $30\ \Omega$
- D. $6.6\ \Omega$

Correct Ans: D

Solution :

Given data:

$$\Rightarrow R_1 = 10\ \Omega$$

$$\Rightarrow R_2 = 20\ \Omega$$

Effective value of resistance (R_{eq}) when two resistors are connected in parallel combin:

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$R_{eq} = \frac{R_1 \times R_2}{R_1 + R_2}$$

$$R_{eq} = \frac{10 \times 20}{10 + 20} = 6.66\ \Omega$$

Q38. The Unit of capacitance is _____

- A. Q/C
- B. C/V
- C. Q/V
- D. C

Correct Ans: B

Solution :

Capacitance:

$\Rightarrow$ Capability of a capacitor to store electrical charge is called capacitance.

$\Rightarrow$ Unit of capacitance is Farad or coulomb/volts and is denoted by F.

$$\Rightarrow Q \propto V, Q = CV, C = \frac{Q}{V} = \frac{\text{coulomb}}{\text{volts}}.$$

Therefore, from the above explanation the correct option is (B) C/V

Q39. Arrange the metals Aluminium, Copper, Iron and Silver in increasing order of thermal conductivity.

- A. Iron, Aluminium, Copper, Silver
- B. Aluminium, Iron, Copper, Silver
- C. Copper, Iron, Silver, Aluminium
- D. Iron, Copper, Aluminium, Silver

Correct Ans: A

Solution :

⇒ The correct order of these metals in increasing order of thermal conductivity is: Iron, Aluminium, Copper, Silver.

Note:

⇒ Iron: Has a thermal conductivity of 50–80 W/m·K

⇒ Aluminum: Has a thermal conductivity of 205–235 W/m·K

⇒ Copper: Has a thermal conductivity of 390–400 W/m·K

⇒ Silver: Has a thermal conductivity of 430 W/m·K

Q40. At what temperature are Celsius and Fahrenheit equal?

- A. $40^{\circ}C$
- B. $40^{\circ}F$
- C. $-40^{\circ}C$ and $-40^{\circ}F$
- D. None of above

Correct Ans: C

Solution :

Relation between Fahrenheit scale and Celsius scale:

$$T_F = \frac{9}{5}T_C + 32$$

Where,

T_F = Temperature on Fahrenheit scale

T_C = Temperature on Celsius scale

Consider $T_C = -40^{\circ}C$, this value put in above equation

$$T_F = \frac{9}{5} \times (-40) + 32$$

$$T_F = -72 + 32$$

$$T_F = -40^{\circ}F$$

Q41. Unit of heat capacity is

- A. J/K
- B. Cal/K
- C. $Cal/^{\circ}C$
- D. All of above

Correct Ans: D

Solution :

Heat Capacity

⇒ It is the ratio of the amount of heat (Q) given to a body to a change in its temperature

$$\Rightarrow H_C = \frac{Q}{\Delta T}$$

⇒ SI Unit: J/K

⇒ Other Unit: Cal/K or $J/^{\circ}C$ or $Cal/^{\circ}C$

Therefore, from the above explanation the correct option is (D) All of above.

Q42. With rising temperatures, the thermal conductivity will be _____

- A. Decreased
- B. Increased
- C. Remain constant
- D. Zero

Correct Ans: The correct answer does not align perfectly with the provided option

Solution :

⇒ The relationship between thermal conductivity and temperature varies depending on material.

Gases: The thermal conductivity of gases rises as the temperature increases.

Liquids: Liquids typically exhibit a decrease in thermal conductivity as temperature increases, although there are some exceptions such as water, which shows an increase initially and then decreases.

Solids: Solids typically exhibit a decrease in thermal conductivity as temperature increases, primarily because higher temperatures lead to heightened lattice vibrations that impede the transfer of heat.

From the above explanation correct options are A and B

Q43. The mathematical formula of specific heat is _____

- A. $C = Q/m\Delta T$
- B. $Q = C/m\Delta T$
- C. $H_c = Q/\Delta L$
- D. $Q = H_c/\Delta L$

Correct Ans: A

Solution :

The mathematical formula for specific heat:

$$C = \frac{Q}{m\Delta T}$$

Where,

C – Specific heat

Q – Heat energy transferred

m – Mass of the substance

ΔT – Change in temperature

Q44. A person is having temperature of $104^{\circ}F$. Convert it in Celsius.

- A. $38^{\circ}C$
- B. $40^{\circ}C$
- C. $-40^{\circ}C$
- D. $-38^{\circ}C$

Correct Ans: B

Solution :

Relation between Fahrenheit scale and Celsius scale:

$$T_F = \frac{9}{5}T_C + 32$$

Where,

T_F = Temperature on Fahrenheit scale

T_C = Temperature on Celsius scale

$$T_F = \frac{9}{5}T_C + 32$$

$$104 = \frac{9}{5}T_C + 32$$

$$104 \times 5 = (9 \times T_C) + (32 \times 5)$$

$$T_C = \frac{(104 \times 5) - (32 \times 5)}{9} = 40^{\circ}C$$

Q45. Which of the following is not a type of electromagnetic wave?

- A. Sound wave
- B. Light wave
- C. Gamma Ray
- D. Micro wave

Correct Ans: A

Solution :

⇒ Sound wave is not a type of electromagnetic wave; it is a mechanical wave that requires medium (such as air, water, or solid material) for propagation.

Q46. Which type of wave is a light wave?

- A. Longitudinal
- B. Transverse
- C. Both (A) and (B)
- D. None of the above

Correct Ans: B

Solution :

⇒ Light waves are classified as non-mechanical type of waves (transverse waves.).

⇒ They do not require any medium for propagation, they can propagate through vacuum.

Note:

Longitudinal wave

⇒ The particles of the medium move parallel to the direction of propagation of wave.

⇒ Examples: Sound waves, etc.

Transverse wave

⇒ The particles of the medium move perpendicular to the direction of propagation of wave.

⇒ Examples: Light waves (electromagnetic waves), Water surface waves, etc.

Therefore, from the above explanation the correct option is (B) Transverse.

Q47. The frequency of sound wave is 2 kHz and its wave length is 2 m . What is the speed of the wave?

- A. 2000 m/s
- B. 400 m/s
- C. 200 m/s
- D. 4000 m/s

Correct Ans: D

Solution :

Given data:

$$\Rightarrow f = 2\text{ kHz} = 2000\text{ Hz}$$

$$\Rightarrow \lambda = 2 m$$

$$v = f \cdot \lambda$$

$$v = 2000 \times 2 = 4000 \text{ m/s}$$

Note:

Frequency:

$\Rightarrow$ It is the number of oscillations or cycles of a wave that occur in a unit of time.

$\Rightarrow$ It is denoted by f

$\Rightarrow$ Unit: hertz (Hz), cycle per second (c/s)

Wavelength:

$\Rightarrow$ It is defined as the distance between two successive points that are in phase.

$\Rightarrow$ It is denoted by λ

$\Rightarrow$ Unit: meter (m) or Angstrom unit ($1 \text{ \AA} = 10^{-10} \text{ m}$)

Q48. _____ is not the characteristics of LASER.

- A. Coherent
- B. Monochromatic
- C. Divergent
- D. High intensity

Correct Ans: C

Solution :

$\Rightarrow$ Divergent is not a characteristic of LASER (Light Amplification by Stimulated Emission Radiation).

Q49. Optical fibre works on the principle of _____

- A. Total internal reflection
- B. Polarization
- C. Refraction
- D. Diffraction

Correct Ans: A

Solution :

$\Rightarrow$ Optical fibre works on the principle of total internal reflection.

Q50. A volume of an auditorium is 6000 m^3 . The total absorption by the surfaces is 1000 O . Then calculate reverberation time.

- A. 9 second
- B. 0.966 second
- C. 90 second
- D. 9.6 second

Correct Ans: B

Solution :

Given data:

$$\Rightarrow V = 6000 \text{ m}^3$$

$$\Rightarrow A = 1000 \text{ O.W.U}$$

$$RT = \frac{0.161 \times 6000}{1000} = 0.966 \text{ seconds}$$

Sabine's formula for reverberation of time:

$$RT = \frac{0.161 \times V}{A}$$

Where,

V is the volume of the room in cubic meters

A is the total absorption in the room in O.W.U .

RT is the reverberation time in seconds

Q51. $\lim_{x \rightarrow 0} \left(\frac{10^x - 5^x}{x} \right) = \underline{\hspace{2cm}}$

- A. $\log_e \frac{1}{2}$
- B. $\log_e 10$
- C. $\log_e 5$
- D. $\log_e 2$

Correct Ans: D

Solution :

$$\lim_{x \rightarrow 0} \left(\frac{10^x - 5^x}{x} \right) = \frac{0}{0}$$

This is of the form $\frac{0}{0}$, so by L'Hospital's rule and differentiate the numerator and denominator

$$= \lim_{x \rightarrow 0} \left(\frac{10^x \log_e 10 - 5^x \log_e 5}{1} \right)$$

$$= \log_e 10 - \log_e 5$$

$$= \log_e 5 + \log_e 2 - \log_e 5$$

$$= \log_e 2$$

Q52. $\lim_{n \rightarrow \infty} \frac{3n^2 - 11n - 13}{(4n - 5)(7 - 6n)} = \underline{\hspace{2cm}}$

- A. $\frac{1}{4}$
- B. $-\frac{1}{8}$
- C. $\frac{1}{6}$
- D. $-\frac{1}{4}$

Correct Ans: B

Solution :

$$\lim_{n \rightarrow \infty} \frac{3n^2 - 11n - 13}{(4n - 5)(7 - 6n)}$$

Divide numerator and denominator by n^2

$$= \lim_{n \rightarrow \infty} \frac{3 - \frac{11}{n} - \frac{13}{n^2}}{(4 - \frac{5}{n})(\frac{7}{n} - 6)}$$

$$= \frac{3 - 0 - 0}{(4 - 0)(0 - 6)}$$

$$= -\frac{1}{8}$$

Q53. If $f(x) = \log_4(x)$ then $f(64) = \underline{\hspace{2cm}}$

- A. 3
- B. -3
- C. 6
- D. $\frac{1}{3}$

Correct Ans: A

Solution :

$$f(x) = \log_4(x) \text{ then}$$

$$f(64) = \log_4(64)$$

$$f(64) = \log_4 4^3$$

$$f(64) = 3\log_4 4$$

$$f(64) = 3$$

Q54. $\frac{d}{dx}(\cos^2 x) = \underline{\hspace{2cm}}$

- A. $\sin 2x$
- B. $-\sin 2x$
- C. $\cos 2x$
- D. $-\cos 2x$

Correct Ans: B

Solution :

$$\Rightarrow \text{For } f(x) = \cos^2 x,$$

$$\Rightarrow f'(x) = \frac{d(\cos^2 x)}{dx} = 2 \cos x \frac{d}{dx} \cos x$$

$$\therefore f'(x) = 2 \cos x (-\sin x) = -\sin(2x)$$

Q55. $\frac{d}{dx}[\log(\sqrt{x^2 - 9})] = \underline{\hspace{2cm}}$

- A. $\frac{1}{\sqrt{x^2 - 9}}$
- B. $\frac{x}{\sqrt{x^2 - 9}}$
- C. $\frac{9}{x^2 - 9}$
- D. $\frac{x}{x^2 - 9}$

Correct Ans: D

Solution :

$$\Rightarrow \text{For } f(x) = \log(\sqrt{x^2 - 9}) = \frac{1}{2} \log(x^2 - 9),$$

$$\Rightarrow f'(x) = \frac{1}{2} \left[\frac{1}{x^2 - 9} \frac{d}{dx} (x^2 - 9) \right]$$

$$\Rightarrow f'(x) = \frac{1}{2} \left[\frac{1}{x^2 - 9} \times 2x \right]$$

$$\Rightarrow f'(x) = \frac{x}{x^2 - 9}$$

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Q56. If $x = \csc \theta$ and $y = \cot \theta$ then $\frac{dy}{dx} = \underline{\hspace{2cm}}$

- A. $\sec \theta$
- B. $\csc \theta$
- C. $\sin \theta$
- D. $\cos \theta$

Correct Ans: A

Solution :

$$\Rightarrow x = \csc \theta \quad \therefore \frac{dx}{d\theta} = -\cot \theta \csc \theta$$

$$\Rightarrow y = \cot \theta \quad \therefore \frac{dy}{d\theta} = -\csc^2 \theta$$

$$\rightarrow \frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{-\csc^2 \theta}{-\cot \theta \csc \theta}$$

$$\therefore \frac{dy}{dx} = \sec \theta$$

Q57. The necessary condition for the function $f(x)$ to be minimum at $x = a$ is $\underline{\hspace{2cm}}$

- A. $f''(a) > 0$
- B. $f''(a) < 0$
- C. $f'(a) > 0$
- D. $f'(a) < 0$

Correct Ans: A

Solution :

Note - This question is canceled by the authority.

$\Rightarrow$ If the function $f(x)$ get that $f''(x) > 0$, then the function will have minimum value.

$\Rightarrow$ If the function $f(x)$ get that $f''(x) < 0$, then the function will have maximum value.

Q58. $\int \cos(10x - 17)dx = \underline{\hspace{2cm}} + c$

- A. $10 \sin(10x - 17)$
- B. $-10 \sin(10x - 17)$
- C. $\frac{1}{10} \sin(10x - 17)$
- D. $-\frac{1}{10} \sin(10x - 17)$

Correct Ans: C

Solution :

$$\Rightarrow \text{Let's consider, } u = 10x - 17 \quad \Rightarrow du = 10dx$$

$$\therefore \int \cos(10x - 17)dx = \frac{1}{10} \int \cos u \, du$$

$$= \frac{1}{10} \sin u + c = \frac{1}{10} \sin(10x - 17) + c$$

Q59. $\int_{-1}^1 \sin^5 x \cos^8 x \, dx = \underline{\hspace{2cm}}$

- A. -1
- B. 0
- C. $\frac{1}{2}$
- D. $\frac{1}{4}$

Correct Ans: B

Solution :

$$\Rightarrow f(x) = \sin^5 x \cos^8 x$$

$$f(-x) = \sin^5(-x) \cos^8(-x) = -\sin^5 x \cos^8 x = -f(x)$$

$\therefore$ The given function $f(x)$ is an Odd function.

$\Rightarrow$ The integral of an odd function over a symmetric interval about the origin is zero.

$$\therefore \int_{-1}^1 f(x)dx = 0$$

Q60. $\int e^x (\csc^2 x - \cot x) dx = \underline{\hspace{2cm}} + c$

- A. $e^x \csc^2 x$
- B. $-e^x \csc^2 x$
- C. $e^x \cot x$
- D. $-e^x \cot x$

Correct Ans: D

Solution :

$$\Rightarrow \text{As rule of integration by parts } \int e^x [f(x) + f'(x)]dx = e^x f(x) + c \quad \dots (i)$$

$$\text{Given : } f(x) = (\csc^2 x - \cot x) = -(\cot x + (-\csc^2 x))$$

$$\rightarrow \text{Consider, } f(x) = \cot x \quad \Rightarrow f'(x) = -\csc^2 x$$

→ From equation (i),

$$\therefore \int e^x (\csc^2 x - \cot x) dx = - \int e^x [(\cot x + (-\csc^2 x))] dx$$

$$\implies \int e^x (\csc^2 x - \cot x) dx = -e^x \cot x + c$$

Q61. $\int \frac{x-3}{x^2-6x+40} dx = \text{---} + c$

- A. $\frac{1}{2} \log |x^2 - 6x + 40|$
- B. $-\frac{1}{2} \log |x^2 - 6x + 40|$
- C. $2 \log |x^2 - 6x + 40|$
- D. $-2 \log |x^2 - 6x + 40|$

Correct Ans: A

Solution :

$\Rightarrow$ Let's consider,

$$u = x^2 - 6x + 40 \implies du = (2x - 6)dx = \frac{1}{2}(x - 3)dx$$

$$\therefore \int \frac{x-3}{x^2-6x+40} dx = \frac{1}{2} \int \frac{1}{u} du$$

$$= \frac{1}{2} \ln |u| + c = \frac{1}{2} \ln |x^2 - 6x + 40| + c$$

Q62. $\log_{\frac{1}{81}} \frac{1}{9} = \text{---}$

- A. 9
- B. $\frac{1}{9}$
- C. $\frac{1}{2}$
- D. $-\frac{1}{2}$

Correct Ans: C

Solution :

$$\text{Given : } \log_{\frac{1}{81}} \frac{1}{9} = \log_{\frac{1}{9^2}} \frac{1}{9}$$

$$= \log_{9^{-2}} 9^{-1} = \frac{-1}{-2} \log_9 9 = \frac{1}{2}$$

Q63. $4^{\log_2 3} + 2^{\log_8 27} = \underline{\hspace{2cm}}$

- A. 9
- B. 10
- C. 11
- D. 12

Correct Ans: D

Solution :

$$\text{Given : } 4^{\log_2 3} + 2^{\log_8 27} = 2^{2\log_2 3} + 8^{\frac{1}{3}\log_8 27}$$

$$= 2^{\log_2 3^2} + 8^{\log_8 27^{\frac{1}{3}}} = 2^{\log_2 9} + 8^{\log_8 3}$$

$$= 9 + 3 = 12$$

Q64. If the mean of the observations 19,15,12,k,8,3 is 11, then k =

- A. 8
- B. 9
- C. 10
- D. 11

Correct Ans: B

Solution :

$$\text{Observations : } 19, 15, 12, k, 8, 3$$

$$\text{Mean} = \frac{19+15+12+k+8+3}{6}$$

$$\Rightarrow 11 = \frac{57+k}{6}$$

$$\Rightarrow 66 = 57 + k$$

$$\therefore k = 9$$

Q65. If $A = \begin{bmatrix} 1 & -1 \\ 1 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix}$ then $(AB)^T = \dots\dots\dots$

- A. $\begin{bmatrix} 3 & -2 \\ 7 & 0 \end{bmatrix}$
- B. $\begin{bmatrix} 3 & 0 \\ -2 & 7 \end{bmatrix}$

C. $\begin{bmatrix} 3 & -2 \\ 0 & 7 \end{bmatrix}$
 D. $\begin{bmatrix} 3 & 0 \\ 7 & -2 \end{bmatrix}$

Correct Ans: B

Solution :

If $A = \begin{bmatrix} 1 & -1 \\ 1 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix}$ then :

$$AB = \begin{bmatrix} 1 & -1 \\ 1 & 2 \end{bmatrix} \cdot \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix}$$

$$AB = \begin{bmatrix} (1)(2) + (-1)(-1) & (1)(1) + (-1)(3) \\ (1)(2) + (2)(-1) & (1)(1) + (2)(3) \end{bmatrix}$$

$$AB = \begin{bmatrix} 2 + 1 & 1 - 3 \\ 2 - 2 & 1 + 6 \end{bmatrix}$$

$$AB = \begin{bmatrix} 3 & -2 \\ 0 & 7 \end{bmatrix}$$

Now, $(AB)^T = \begin{bmatrix} 3 & -2 \\ 0 & 7 \end{bmatrix}^T$

$$(AB)^T = \begin{bmatrix} 3 & 0 \\ -2 & 7 \end{bmatrix}$$

Q66.

If $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 3 \\ 1 & 1 & 0 \end{bmatrix}$ then $|A| = \dots\dots\dots$.

- A. -3
- B. 3
- C. -4
- D. 4

Correct Ans: D

Solution :

$$\text{For } A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 3 \\ 1 & 1 & 0 \end{bmatrix}$$

$$|A| = 1((1)(0) - (3)(1)) - 2((2)(0) - (3)(1)) + 1((2)(1) - (1)(1))$$

$$|A| = 1(0 - 3) - 2(0 - 3) + 1(2 - 1)$$

$$|A| = -3 + 6 + 1$$

$$|A| = 4$$

Q67. If $A = \begin{bmatrix} 2 & -1 \\ 3 & 1 \end{bmatrix}$ then $A^{-1} = \dots\dots\dots$.

A. $\frac{1}{5} \begin{bmatrix} 1 & 1 \\ -3 & 2 \end{bmatrix}$

B. $\frac{1}{5} \begin{bmatrix} 2 & 1 \\ -3 & 1 \end{bmatrix}$

C. $\frac{1}{5} \begin{bmatrix} 1 & -1 \\ 3 & 2 \end{bmatrix}$

D. $\frac{1}{5} \begin{bmatrix} -2 & 3 \\ -1 & -1 \end{bmatrix}$

Correct Ans: A

Solution :

$$\text{If } A = \begin{bmatrix} 2 & -1 \\ 3 & 1 \end{bmatrix}$$

Adjoint of a Matrix A (two by two) can be found by changing values of diagonal element changing signs of off-diagonal elements.

$$\text{Adj}(A) = \begin{bmatrix} 1 & 1 \\ -3 & 2 \end{bmatrix}$$

$$\text{Determinant } |A| = \begin{vmatrix} 2 & -1 \\ 3 & 1 \end{vmatrix} = 5$$

Inverse of matrix A is given by:

$$(A)^{-1} = \frac{\text{Adj}(A)}{|A|} = \frac{\begin{bmatrix} 1 & 1 \\ -3 & 2 \end{bmatrix}}{|A|}$$

$$(A)^{-1} = \frac{1}{5} \begin{bmatrix} 1 & 1 \\ -3 & 2 \end{bmatrix}$$

Q68. If $A = \begin{bmatrix} 2 & -1 \\ 1 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 2 & 3 \end{bmatrix}$ then $3B - 2A = \dots\dots\dots$.

- A. $\begin{bmatrix} 4 & -3 \\ -1 & 6 \end{bmatrix}$
- B. $\begin{bmatrix} 4 & -3 \\ -1 & -6 \end{bmatrix}$
- C. $\begin{bmatrix} -1 & 2 \\ 4 & -9 \end{bmatrix}$
- D. $\begin{bmatrix} -1 & 2 \\ 4 & 9 \end{bmatrix}$

Correct Ans: D

Solution :

If $A = \begin{bmatrix} 2 & -1 \\ 1 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 2 & 3 \end{bmatrix}$ then :

$$\Rightarrow 3B - 2A = 3 \times \begin{bmatrix} 1 & 0 \\ 2 & 3 \end{bmatrix} - 2 \times \begin{bmatrix} 2 & -1 \\ 1 & 0 \end{bmatrix}$$

$$\Rightarrow 3B - 2A = \begin{bmatrix} 3 & 0 \\ 6 & 9 \end{bmatrix} - \begin{bmatrix} 4 & -2 \\ 2 & 0 \end{bmatrix}$$

$$\Rightarrow 3B - 2A = \begin{bmatrix} -1 & 2 \\ 4 & 9 \end{bmatrix}$$

Q69. $\sin(225^\circ) = \dots\dots\dots$.

- A. $\frac{-1}{\sqrt{2}}$
- B. $\frac{1}{\sqrt{2}}$
- C. $\frac{\sqrt{3}}{2}$
- D. $\frac{-1}{2}$

Correct Ans: A

Solution :

$$\begin{aligned}
 & \sin(225^\circ) \\
 &= \sin(180^\circ + 45^\circ) \\
 &= -\sin 45 \\
 &= -\frac{1}{\sqrt{2}}
 \end{aligned}$$

Q70. If $\tan \theta = -\frac{12}{5}$, $\frac{3\pi}{2} < \theta < 2\pi$ then $\cos \theta = \dots\dots\dots$.

- A. $\frac{-12}{13}$
- B. $\frac{12}{13}$
- C. $\frac{5}{13}$
- D. $\frac{-5}{13}$

Correct Ans: C

Solution :

$$\sec^2 \theta - \tan^2 \theta = 1$$

$$\sec^2 \theta = 1 + \tan^2 \theta$$

$$\sec^2 \theta = 1 + \left(\frac{12}{5}\right)^2$$

$$\sec^2 \theta = 1 + \frac{144}{25}$$

$$\sec^2 \theta = \frac{169}{25}$$

$$\sec \theta = \frac{13}{5}$$

$$\implies \cos \theta = \frac{5}{13}$$

Q71. $\cos^{-1}\left(-\frac{\sqrt{3}}{2}\right) = \dots\dots\dots$.

- A. $\frac{2\pi}{3}$
- B. $\frac{-\pi}{6}$
- C. $\frac{5\pi}{3}$
- D. $\frac{5\pi}{6}$

Correct Ans: D

Solution :

Note - This question is canceled by the authority.

In $\cos^{-1}\left(-\frac{\sqrt{3}}{2}\right)$, where $\left(-\frac{\sqrt{3}}{2} < 0\right)$

$\rightarrow$ It means $\cos^{-1}\left(-\frac{\sqrt{3}}{2}\right)$ lies in $2^{nd} / 3^{rd}$ quadrant.

$$\text{Now } \cos^{-1}\left(\frac{\sqrt{3}}{2}\right) = \frac{\pi}{6}$$

$$\text{So, } \cos^{-1}\left(-\frac{\sqrt{3}}{2}\right) = \left(\pi + \frac{\pi}{6}\right) \text{ or } \left(\pi - \frac{\pi}{6}\right)$$

$$\cos^{-1}\left(-\frac{\sqrt{3}}{2}\right) = \left(\frac{7\pi}{6}\right) \text{ or } \left(\frac{5\pi}{6}\right)$$

Q72. If $|\vec{x} \times \vec{y}| = |\vec{x}| = |\vec{y}| = \sqrt{2}$ then $(\angle \vec{x}, \vec{y}) = \dots\dots\dots$.

- A. $\frac{\pi}{3}$
- B. $\frac{\pi}{4}$
- C. $\frac{\pi}{6}$
- D. $\frac{\pi}{2}$

Correct Ans: B

Solution :

$$|\vec{x} \times \vec{y}| = |\vec{x}| |\vec{y}| \sin \theta_{xy}$$

It is given that $|\vec{x} \times \vec{y}| = |\vec{x}| = |\vec{y}| = \sqrt{2}$

$$\Rightarrow \sqrt{2} = \sqrt{2} \sqrt{2} \sin \theta_{xy}$$

$$\Rightarrow \sin \theta_{xy} = \frac{1}{\sqrt{2}}$$

$$\Rightarrow \theta_{xy} = \frac{\pi}{4}$$

Q73. If vectors $(2, 3, a)$ and $(3, -1, 4)$ are perpendicular then $a = \dots\dots\dots$.

- A. $\frac{-3}{4}$
- B. $\frac{3}{4}$
- C. $\frac{4}{3}$

D. $\frac{-4}{3}$

Correct Ans: A

Solution :

If vectors $(2, 3, a)$ and $(3, -1, 4)$ are perpendicular then $(2, 3, a) \cdot (3, -1, 4) = 0$

$$\implies (2, 3, a) \cdot (3, -1, 4) = 0$$

$$6 - 3 + 4a = 0$$

$$3 + 4a = 0$$

$$a = \frac{-3}{4}$$

Q74. If the centre of the circle $x^2 + y^2 - 6my + 11 = 0$ is $(0, -3)$ then $m = \dots\dots\dots$.

- A. 2
- B. -2
- C. 1
- D. -1

Correct Ans: D

Solution :

For the general equation of circle $x^2 + y^2 + 2gx + 2fy + c = 0$, the center of the circle $(-g, -f)$

By comparing it with given equation we get:

$$g = 0, f = -3m$$

the center of the circle is given as $(0, -3)$:

$$\implies -f = 3m = -3$$

$$\implies m = -1$$

Q75. Equation of the line passing through the points $(1, -1)$ and $(-1, 1)$ is

- A. $2x + y - 1 = 0$
- B. $x + 2y - 1 = 0$
- C. $x + y = 0$
- D. $x + 2y + 1 = 0$

Correct Ans: C

Solution :

⇒ Two-point form of line joining two points (x_1, y_1) and (x_2, y_2) is given by:

$$\frac{y-y_1}{y_2-y_1} = \frac{x-x_1}{x_2-x_1}$$

By placing given two points $(1, -1)$ and $(-1, 1)$, above equation can be written as:

$$\Rightarrow \frac{y-(-1)}{1-(-1)} = \frac{x-1}{(-1)-1}$$

$$\Rightarrow \frac{y+1}{2} = \frac{x-1}{-2}$$

$$\Rightarrow -2y - 2 = 2x - 2$$

$$\Rightarrow 2x + 2y = 0$$

$$\Rightarrow x + y = 0$$

Q76. Which of the following is not a part of the business letter?

- A. Photo
- B. Salutation
- C. Date
- D. Signature

Correct Ans: A

Solution :

⇒ A photo is not typically a part of a standard business letter. Therefore, among the options provided, "Photo" is not a part of the business letter format.

⇒ The other elements listed (Salutation, Date, Signature) are commonly included in a business letter.

Q77. A letter is incomplete and worthless without_____.

- A. Signature
- B. Postscript
- C. Enclosure
- D. Photo

Correct Ans: A

Solution :

⇒ A letter is incomplete and worthless without a signature.

⇒ The signature is a crucial element that indicates authenticity and authorship, confirm sender's identity and agreement with the content of the letter. Therefore, "Signature" is correct answer.

Q78. Give the full form of C.O.D.

- A. Cash on delay
- B. Cash on delivery
- C. Cash on order
- D. Cash on distance

Correct Ans: B

Solution :

⇒ The full form of C.O.D. is "Cash on Delivery." It refers to a payment method where payment is made upon the delivery of goods rather than in advance.

Q79. Hurrah! I have passed the examination. Identify the interjection.

- A. Examination
- B. Passed
- C. Hurrah!
- D. The whole sentence.

Correct Ans: C

Solution :

⇒ Interjections are words or phrases used to express emotions or sentiments and are followed by an exclamation mark. In this case, "Hurrah!" expresses joy or excitement at passing the examination.

Q80. The dog is sitting _____ the cot. Select appropriate option.

- A. under
- B. into
- C. between
- D. up

Correct Ans: A

Solution :

⇒ The appropriate option to complete the sentence is "under." Therefore, the sentence be: "The dog is sitting under the cot." This option indicates the location of the dog in relation to the cot.

Q81. The Sun _____ in the East. Select appropriate option.

- A. raise
- B. rises
- C. rise
- D. raised

Correct Ans: B

Solution :

⇒ The appropriate option to complete the sentence is "rises." Therefore, the complete sentence is: "The Sun rises in the East." This option correctly conjugates the verb "rise" to agree with the subject "Sun" and the present tense of the sentence.

Q82. Your family _____ a happy family. Select appropriate option.

- A. are
- B. is
- C. has
- D. have

Correct Ans: B

Solution :

⇒ In this sentence, "Your family" is the subject, and it refers to a singular entity, even though it consists of multiple individuals. "Is" is the appropriate verb to use with a singular subject in the present tense.

Q83. Do you have _____ doubt in this question paper? Select appropriate option.

- A. some
- B. any
- C. many
- D. much

Correct Ans: B

Solution :

⇒ The appropriate option to complete the sentence is "any." Therefore, the complete sentence is: "Do you have any doubt in this question paper?"

⇒ The word "any" is used in questions to inquire about the existence or presence of something, especially when the answer could be affirmative or negative. In this context, it is asking if there is doubt in the question paper, implying that the speaker is open to the possibility of an answer.

Q84. Select the correct spelling.

- A. Playwright
- B. Playwrite
- C. Playright
- D. Playrite

Correct Ans: A

Solution :

⇒ A "playwright" is a person who writes plays. The term "playwright" combines the word referring to a dramatic work meant for performance on a stage, and "wright," which historically meant a craftsman or someone who creates something. So, a playwright is essentially someone who crafts or creates plays.

Q85. Select the correct spelling.

- A. Enterpeneur
- B. Entrepreneur
- C. Enterpreneur
- D. Entrepreneur

Correct Ans: B

Solution :

⇒ The correct spelling is "Entrepreneur."

Q86. Choose the correct grammatical sentence.

- A. People were coming, going and ignorant him.
- B. People were coming, going and ignoring him.
- C. People were coming, going and ignoring him
- D. People were coming, going and ignored him.

Correct Ans: B

Solution :

⇒ The correct grammatical sentence is: "People were coming, going and ignoring him."

⇒ This sentence uses the correct verb form "ignoring" to match the progressive tense of the sentence ("were coming, going"). The verb "ignoring" is in the present participle form, which is appropriate for indicating ongoing action in the past tense.

Q87. Choose the correct grammatical sentence.

- A. I like drawing my coworkers and my cat.
- B. I Like drawing, my coworkers and my cat.
- C. I like drawing, my coworkers, and, my cat.
- D. I, Like, drawing, my coworkers, and, my, cat.

Correct Ans: B

Solution :

Note - This question is canceled by the authority.

⇒ The grammatically correct sentence is:

I like drawing, my coworkers and my cat.

⇒

Here's why the other options are incorrect:

→ I Like drawing my coworkers and my cat. - No proper use of comma.

→ I like drawing, my coworkers, and, my cat. - This one has unnecessary commas.

→ I, Like, drawing, my coworkers, and, my, cat. - This one uses unnecessary commas and capitalization.

⇒ In option B, sentence consists of a simple subject ("I"), a verb ("like"), and a direct object ("drawing my coworkers and my cat"). This structure is clear, concise, and grammatically correct.

Q88. Choose the correct grammatical sentence.

- A. The ship was wrecked, and every man, woman and child have drowned.
- B. The ship was wrecked, and every man, woman, and child are drowned.
- C. The ship was wrecked, and every man, woman and child is drowned.
- D. The ship was wrecked, and every man, woman and child was drowned.

Correct Ans: D

Solution :

⇒ The correct grammatical sentence is: "The ship was wrecked, and every man, woman and child was drowned."

⇒ In this sentence: "The ship was wrecked" is in the past tense, which matches the action of the shipwreck.

"Every man, woman, and child" is a collective noun that refers to a singular group, so it takes a singular verb form, "was drowned," to match.

⇒ Therefore, the correct form is "was drowned," making the sentence grammatically correct.

Q89. Read the following passage carefully and answer the questions: (Q.No. 08)

The science of fireworks is technically called 'pyrotechnics' from the Greek word 'pyr' meaning fire and 'technics' meaning an art. Pyrotechnics includes not only fireworks but also other devices that use similar materials and principles, from safety matches that we use every day to solid fuel rocket boosters of space shuttle. The household match is considered a special pyrotechnic device, as all the pyrotechnic effects—heat, smoke, light, gas and sound—are produced by it.

Some historians say that 'black powder', the basic material used in fireworks, was invented

India, Shukranti, written more than two thousand years ago, has references to weapons to guns and projectile weapons. However, the Chinese are generally considered the pioneers of pyrotechnics. They are said to have developed, black powder more than one thousand years ago. It took at least two thousand years for the knowledge to spread to the West, and it was in 1242 that an English Monk, Roger Bacon, revealed the formula for "black powder". He considered it such a dangerous substance that he wrote of it in a code language.

Ques. As per some historians 'Black Powder was invented in _____.

- A. India
- B. China
- C. Greek
- D. None of the above

Correct Ans: A

Solution :

⇒ Black Powder was invented in India.

Q90. Roger Bacon was _____.

- A. a Chinese writer
- B. an Indian historian
- C. an English Monk
- D. a Greek philosopher

Correct Ans: C

Solution :

⇒ Roger Bacon was an English Monk.

Q91. Who wrote about 'Black Powder in code language?

- A. Shukranti
- B. Roger Bacon
- C. Chinese
- D. Greek

Correct Ans: B

Solution :

⇒ Roger Bacon wrote about 'Black Powder' in code language.

Q92. When was Shukranti written?

- A. One thousand years ago
- B. In 1242
- C. Two thousand years ago
- D. Modern times

Correct Ans: C

Solution :

⇒ Shukranti was written "Two thousand years ago."

Q93. Pyrotechnics word is from

- A. Indian language
- B. Greek language
- C. Chinese language
- D. None of the above

Correct Ans: B

Solution :

⇒ The word "pyrotechnics" is from the Greek language.

Q94. Verbal communication can be in _____ form.

- A. written
- B. oral and written
- C. oral
- D. none of these

Correct Ans: B

Solution :

⇒ Verbal communication refers to using words to convey information. This can be done both speaking (oral) and writing.

Q95. Tone of voice, voice quality, style of speaking, stress, intonation are part of _____

- A. body language
- B. sign language
- C. paralanguage
- D. code language

Correct Ans: C

Solution :

⇒ Tone of voice, voice quality, style of speaking, stress, and intonation are part of "paralanguage."

⇒ Paralanguage refers to the non-verbal elements of speech, such as tone, pitch, volume, rhythm, which convey meaning alongside spoken words.

Q96. The process of communication is a _____.

- A. two way
- B. three way
- C. one way
- D. single way

Correct Ans: A

Solution :

⇒ The process of communication is a "two way" process. This implies that communication involves both sending and receiving messages between two or more parties.

Q97. "During the process of communication at every stage, there is hindrance which disturbs communication process." Such hindrance is known as

- A. barrier
- B. noise
- C. sender
- D. receiver

Correct Ans: A

Solution :

⇒ The hindrance that disturbs the communication process at every stage is known as a "Barrier." Noise can be considered as one of the barrier.

Q98. Our physical appearance is an example of _____ communication.

- A. oral
- B. written
- C. verbal
- D. non-verbal

Correct Ans: D

Solution :

⇒ Our physical appearance is an example of "non-verbal" communication. Non-verbal communication includes gestures, facial expressions, body language, posture, and other aspects of appearance that convey messages without the use of words.

Q99. What is written at the top of a business letter on the left hand side.

- A. Sender's address
- B. Receiver's address
- C. Date
- D. Salutation

Correct Ans: A

Solution :

⇒ At the top of a business letter on the left-hand side, you typically find the "Sender's address." This includes the sender's name, street address, city, state, and postal code.

Q100. Where is the salutation placed in a business letter?

- A. Just above the date
- B. Just below the date
- C. Along with the subject
- D. None of these

Correct Ans: D

Solution :

⇒ The salutation in a business letter is typically placed "Just above the body of the letter." is the greeting that addresses the recipient of the letter.

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